simon-5502-05-slides

2025-02-18

Topics to be covered

- What you will learn
  - Mathematical model of interactions
  - Interactions in multi-factor analysis of variance
  - Other examples of interactions
  - Interactions in analysis of covariance
  - Interactions in multiple linear regression

Mathematical model, 1

- $Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk}$
  - i=1,…,a j=1,…,b, k=1,…,n
- If 1 is the reference category
  - $\alpha_1 = 0$
  - $\beta_1 = 0$
  - $(\alpha\beta)_{1j} = 0$
  - $(\alpha\beta)_{i1} = 0$

Speaker notes

You may see papers or books that present the mathematical model for an interaction. The model I present is a balanced model with the first category having levels one through a, the second category having levels one through b and for each combination of categories there are n observations.

If you set the first level as the reference category for each category, then you need to set some of these parameters to zero.

Interpretation

- μ, estimated average outcome for both A, B at reference level
- α average change from reference level for A holding B fixed at reference level
- β average change from reference level for B holding A fixed at reference level
- $\alpha\beta$ average deviation from parallelism

Speaker notes

The interpretation of these parameters is a bit different when you have an interaction. The overall mean, mu, is the estimated average outcome when A and B are both at the reference level. This much has not changed.

The interpretation of the alpha values is different, but in a subtle way. It represents the average change when you switch A from the reference level to one of the other reference levels, holding B constant but constant at the reference level for B.

Similarly, beta represents the average change when you switch B from the reference level to one of the other reference levels, holding B constant at the reference level for B.

The interaction term, alpha-beta, is the degree to which a particular combination of non-reference levels for A and B differ from parallelism. Parallelism means a model with no interaction. You can think of the interaction term as the degree of super-additivity (if the term is positive) or sub-additivity (if the term is negative).

Mathematical model, 2

- $SS_A = \Sigma_i nb(\bar{Y}_{i..} - \bar{Y}_{...})^2$
- $SS_B = \Sigma_j na(\bar{Y}_{.j.} - \bar{Y}_{...})^2$
- $SS_{AB} = \Sigma_i \Sigma_j n(\bar{Y}_{ij.} - \bar{Y}_{i..} - \bar{Y}_{.j.} + \bar{Y}_{...})^2$
- $SS_E = \Sigma_i \Sigma_j \Sigma_k (Y_{ijk} - \bar{Y}_{ij.})^2$
- $SS_T = \Sigma_i \Sigma_j \Sigma_k (Y_{ijk} - \bar{Y}_{...})^2$

Speaker notes

The dot notation may be a bit confusing until you get used to it, but $\bar{Y}_{i..}$ is the average within the ith group, averaging across the subscripts j and k. $\bar{Y}_{.j.}$ is the average within the ith group, averaging across the subscripts i and k. $\bar{Y}_{ij.}$ is the average within the combination of the ith group and the jth group, averaging across the subscript k. Finally, $\bar{Y}_{...}$ is an overall mean and the average across all three subscripts.

Test for an interaction

- SS_{AB} has (a-1)(b-1) degrees of freedom
- SS_E has ab(n-1) degrees of freedom
- Accept H_0 if $F = \frac{MS_{AB}}{MS_E}$ is close to one
  - In R, fit a model without an interaction
  - Compare to a model with interaction
  - Using the anova function

Speaker notes

The formal test for an interaction uses an F ratio and you accept the null hypothesis if that F ratio is close to one. You would reject the null hypothesis if the F ratio is much larger than one.

It is not easy to get R to display all the sums of squares and mean squares that I defined above. Instead, compute two models-one without an interaction and one with an interaction. Compare those two models using the anova function.

Break #1

- What you have learned
  - Mathematical model of interactions
- What's coming next
  - Interactions in multi-factor analysis of variance

Hypothetical applications

- Study with two drugs and two placebos
- Four treatment options
  - Placebo plus placebo
  - Drug A plus placebo
  - Placebo plus Drug B
  - Drug A plus Drug B

Speaker notes

Consider a hypothetical case. This is a fairly common research scenario where you are looking at the combination effect of two drugs, A and B. The comparison is to a placebo for each drug. This leads to four choices, placebo for A plus a placebo for B, active drug A plus a placebo for drug B, placebo for A plus active drug for B, and finally active drug A plus active drug B.

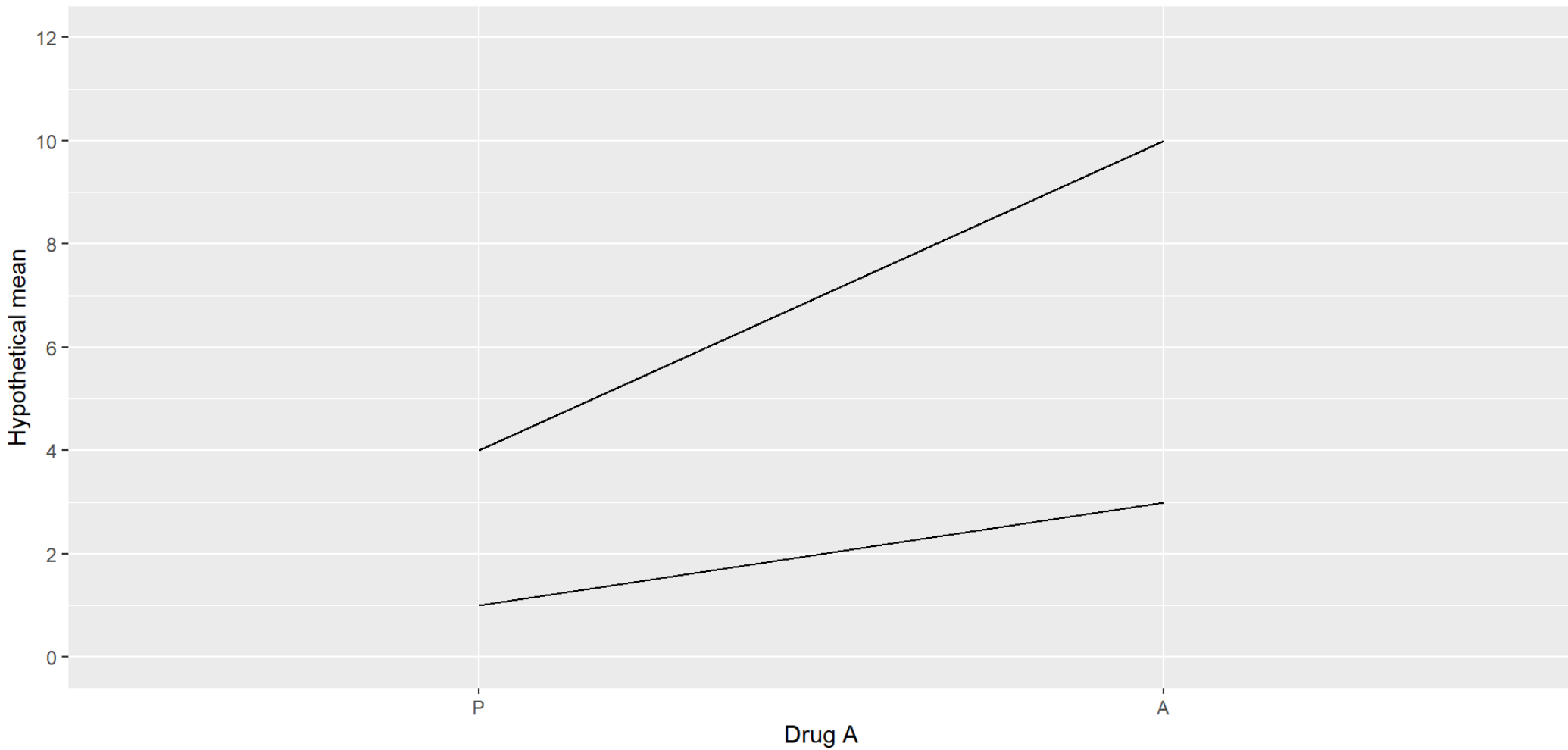
Listing of hypothetical data

```
# A tibble: 8 × 3
  a     b         y
  <fct> <fct> <dbl>
1 P     P       0.9
2 P     P       1.1
3 A     P       2.9
4 A     P       3.1
5 P     B       3.9
6 P     B       4.1
7 A     B       9.9
8 A     B      10.1
```

Speaker notes

This is what the data might look like if each combination had two observations.

Line plot of hypothetical data, 1



Steve Simon, 2025-02-18, CC0

Speaker notes

This is what the interaction plot looks like for this hypothetical data. Notice the lack of parallelism.

Create indicator variables

```
# A tibble: 8 × 6
  a     b       i_a   i_b  i_ab     y
  <fct> <fct> <dbl> <dbl> <dbl> <dbl>
1 P     P         0     0     0   0.9
2 P     P         0     0     0   1.1
3 A     P         1     0     0   2.9
4 A     P         1     0     0   3.1
5 P     B         0     1     0   3.9
6 P     B         0     1     0   4.1
7 A     B         1     1     1   9.9
8 A     B         1     1     1  10.1
```

Speaker notes

Here is what the indicator variables look like. Notice that the indicator for the interaction is just the product of the indicator for drug A and the indicator for drug B.

Interaction model, 1

- `lm(y=a+b+a:b, data=hyp)`
- `lm(y=a*b, data=hyp)`
- `lm(y=i_a+i_b+i_ab, data=hyp_1)`

Speaker notes

You can fit the interaction term by putting a colon between a and b. An asterisk between a and b means to fit the main effect of a, the main effect of b and the interaction between a and b.

Alternately, you can create indicator variables and use them directly.

Estimates from the hypothetical data

```
# A tibble: 4 × 2
  term        estimate
  <chr>          <dbl>
1 (Intercept)     1.00
2 i_a             2
3 i_b             3
4 i_ab            4
```

Speaker notes

Here are the estimates from the hypothetical data. The following slides will show you the proper interpretations of the estimates.

Interaction model, 2

- (Intercept): estimated average outcome with two placebos
- i_a: estimated average change due to drug A alone
- i_b: estimated average change due to drug B alone
- i_ab: estimated synergistic (or antagonistic) effect of both drugs

Speaker notes

The intercept is the estimated average value of Y when all of the indicator variables are equal to zero. This is the average in the P P group. The slope associated with i_a is the estimated average change when moving from placebo to drug A, holding drug B constant. The slope associated with i_b is the estimated average change when moving from placebo to drug B, holding drug A constant. The slope associated with i_ab is the interaction, a measure of how much more the effect of the two drugs combined compared to the effect of drug A along plus drug B alone.

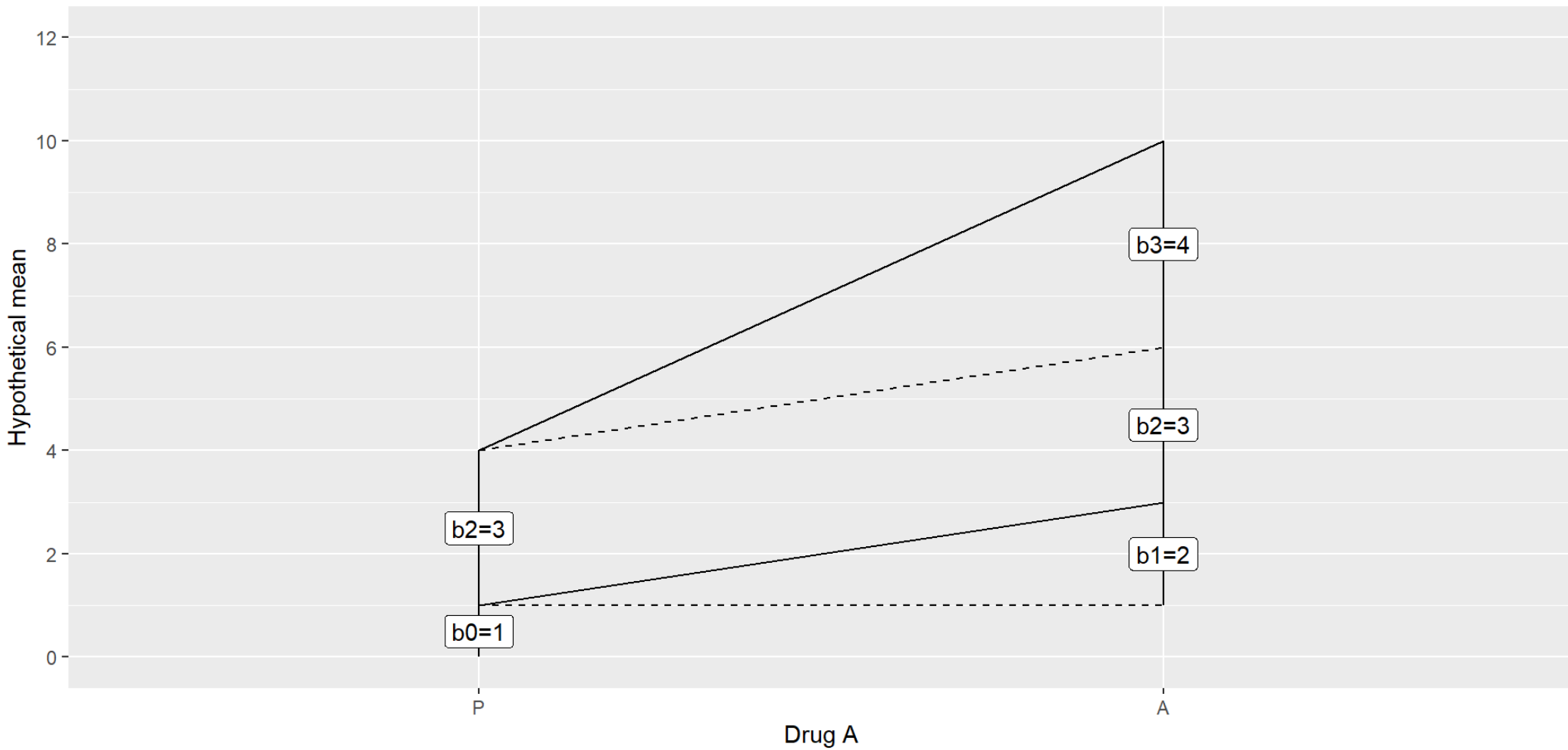
Interaction model, 3

- `P P mean = 1`
- `A P mean = 1 + 2`
- `P B mean = 1     + 3`
- `A B mean = 1 + 2 + 3 + 4`

Speaker notes

The P P mean, the mean for both placebos, is the estimated intercept. The A P mean, the mean for drug A plus placebo, is the estimated intercept plus the estimate for the indicator for drug A. The P A mean, the mean for placebo plus drug B, is the estimated intercept plus the estimate for the indicator for drug B. The A B mean, the mean for drug A plus drug B, would be, if there was no interaction, the estimated intercept plus the estimate for the indicator for drug A plus the indicator for drug B. No interaction means that the combined effect of the two drugs is additive. With an interaction, you have to add in an extra term, the indicator for the interaction. This is a measure of the degree of super-additivity or sub-additivity. It is a measure of the degree of synergy or antagonism. In this case there is super-additivity or a lot of synergy.

Line plot of hypothetical data, 2



Steve Simon, 2025-02-18, CC0

Speaker notes

Here is the interaction plot, showing the impact of the individual estimates.

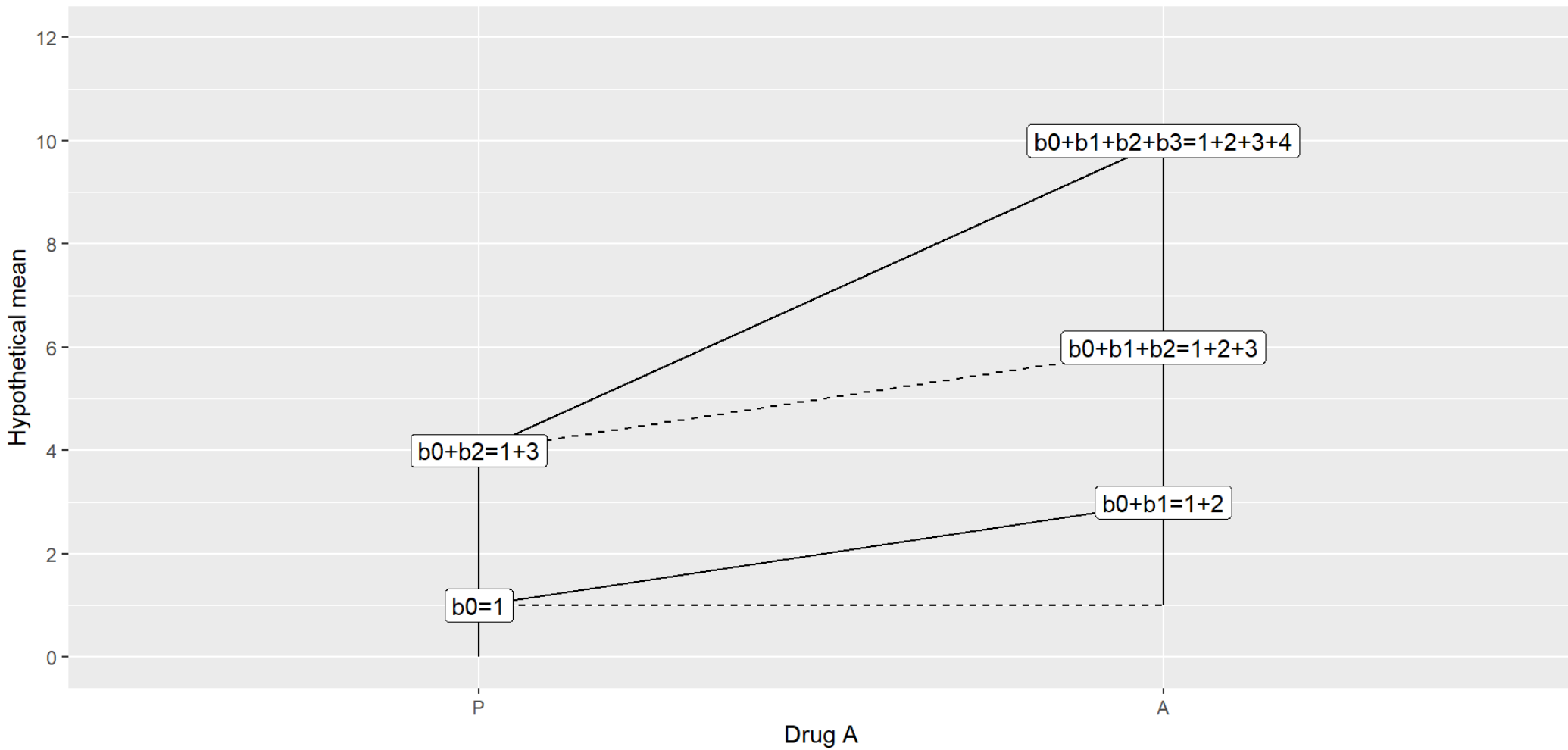
The intercept is 1 showing that the double placebo has an estimated average outcome of 1.

The estimate for the indicator for A is the change when you move from the double placebo to an active version of A while keeping B as the placebo. The individual effect of drug A by itself is an increase of 2 units on average in the outcome.

The estimate for the indicator for B is the change when you move from the double placebo to an active version of B while keeping A as the placebo. The individual effect of drug B bt itself is an increase of 3 units on average in the outcome.

The estimate for the indicator for the AB interaction is the degree to which the lines depart from parallelism.

Line plot of hypothetical data, 3



Steve Simon, 2025-02-18, CC0

Speaker notes

This alternate version of the plot shows how to compute the estimated average for each of the combinations: double placebo, A alone, B alone, and A and B commined.

The estimated average for the double placebo is the estimated intercept = 1.

The estimated average for drug A alone is the estimated intercept plus the estimate for the indicator for drug A = 1 + 2.

The estimated average for drug B alone is the estimated intercept plus the estimate for the indicator for drug B = 1 + 3.

The estimated average for both active drugs is the estimated intercept plus the estimate for both indicators plus the estimate for the indicator of interaction = 1 + 2 + 3 + 4. If there were no interaction, the combined effect would only = 1 + 2 + 3.

Tracking a moving target

- gender (M or F)
- child (Yes or No)
- trial4 (tracking time in seconds)

Speaker notes

Let’s see what an interaction looks like with some more realistic data. Volunteers were given a test of motor skills where they tried to track a moving target. The outcome is the amount of time in seconds that the subjects were successful in tracking. There were both children and adults of both genders in the dataset.

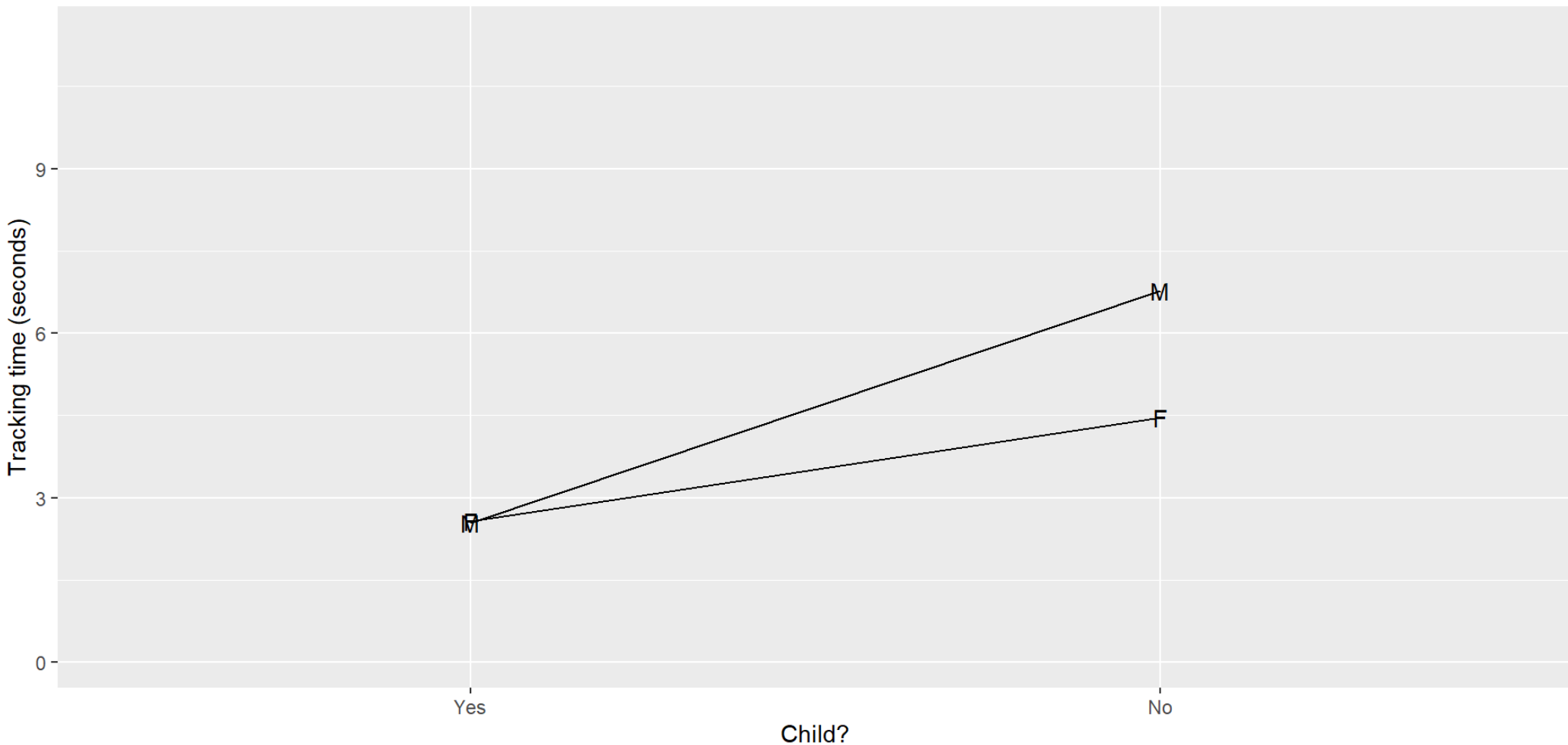
Means for each combination of gender and child

```
# A tibble: 4 × 3
  child gender mean_trial4
  <fct> <chr>        <dbl>
1 Yes   F             2.57
2 Yes   M             2.54
3 No    F             4.46
4 No    M             6.77
```

Speaker notes

Here are the means and standard deviations for the four combinations: female child, male child, female adult, male adult. Notice that there is almost no diference between male and female children. There is a difference in adults and adults of both genders did better than the children.

Interaction plot, 1

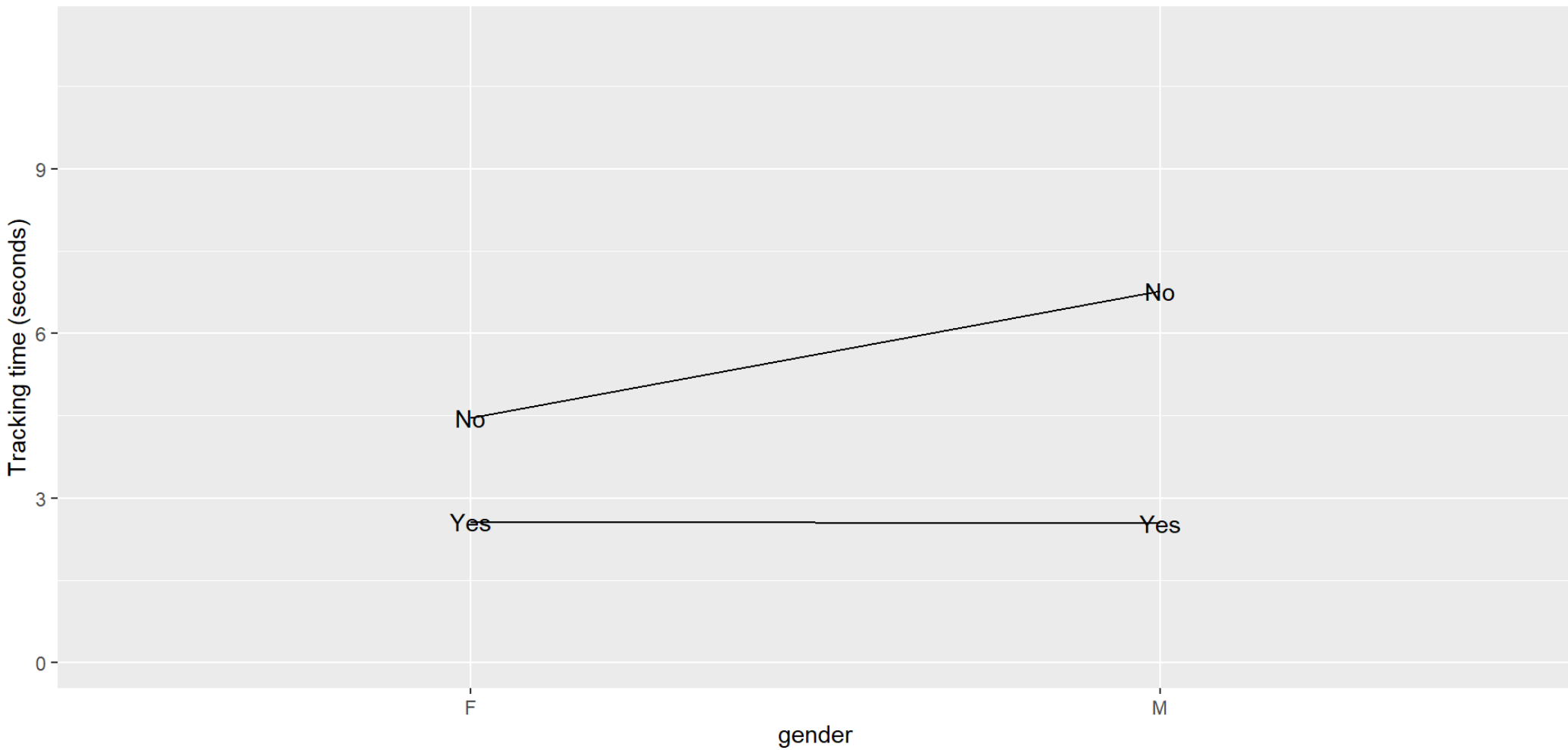


Simon, 2026-02-05

Speaker notes

Here is the interaction plot, showing a line for females and a line for males.

Interaction plot, 2



Simon, 2026-02-10

Speaker notes

Here is the alternate interaction plt, showing a line for children and a line for adults. I like this interaction plot a bit better, but both are good.

The child line is flat, showing no gender effect in children. The adult line has a positive slope showing that adult males did a better job on averag in tracking the target than adult females. The separation between the two lines shows that adults of either gender were better than the children of either gender.

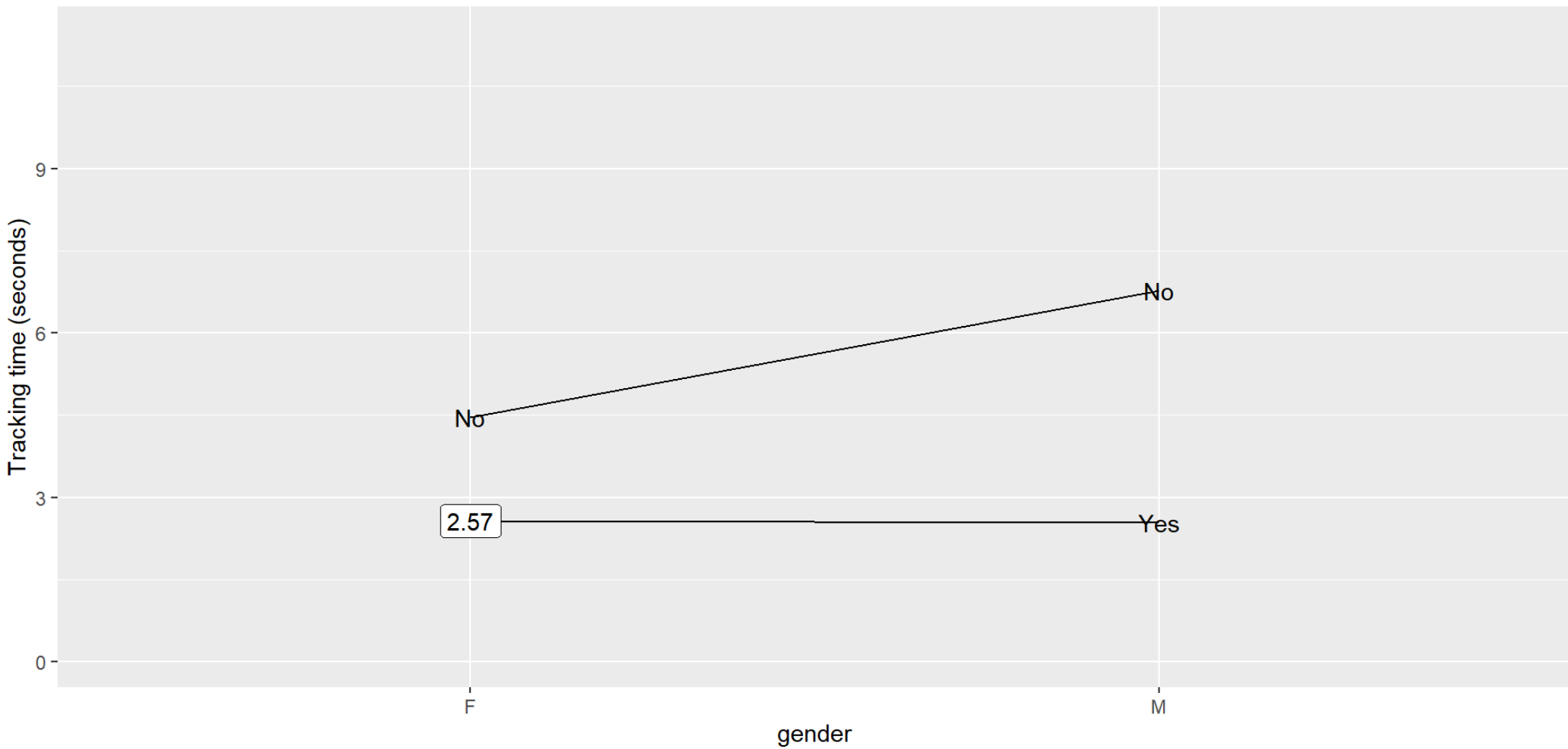
Regression estimates for the interaction model

```
# A tibble: 4 × 5
  term            estimate std.error statistic p.value
  <chr>              <dbl>     <dbl>     <dbl> <glue>
1 (Intercept)       2.57       0.382    6.73   p < 0.001
2 genderM          -0.0343     0.516   -0.0666 p = 0.947
3 childNo           1.89       0.654    2.89   p = 0.005
4 genderM:childNo   2.35       0.815    2.88   p = 0.005
```

Speaker notes

Here are the regression estimates. I will show where these estimates appear within the interaction plot.

intercept

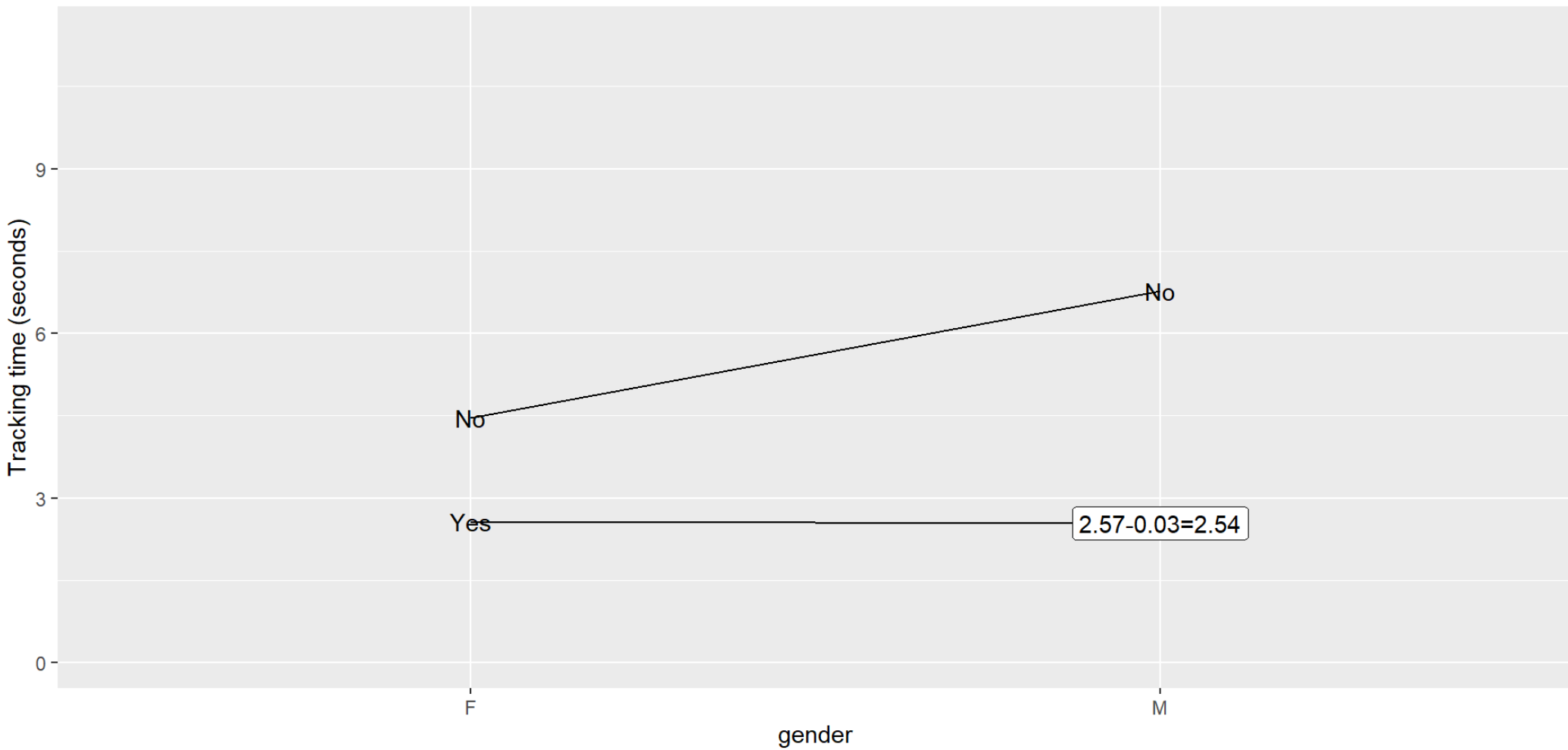


Simon, 2026-02-10

Speaker notes

The intercept is the estimated average tracking time for the two reference categories (female and child).

intercept + genderMale

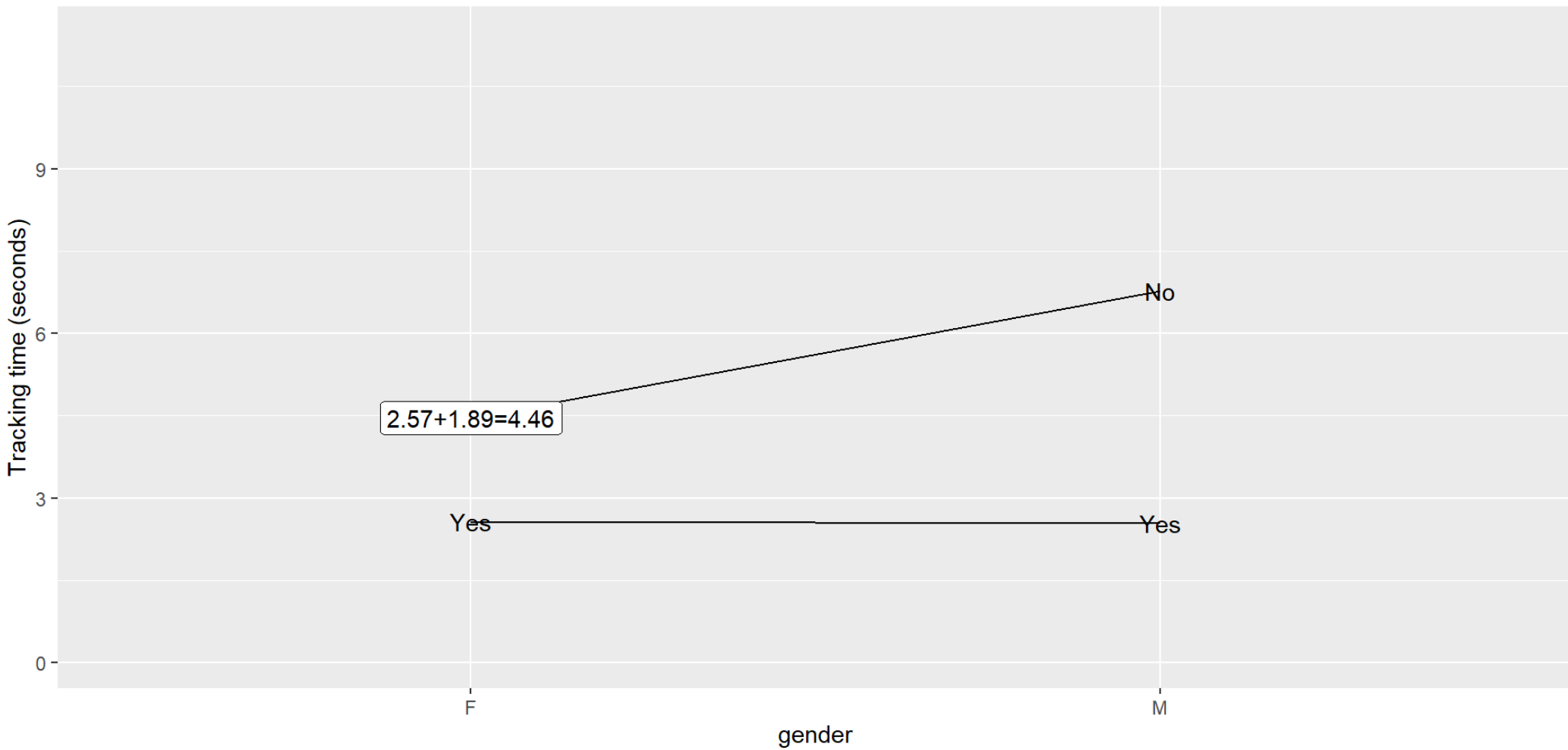


Simon, 2026-02-10

Speaker notes

Add the genderMale term to the intercept to get the estimated average tracking time for male children.

intercept + childNo

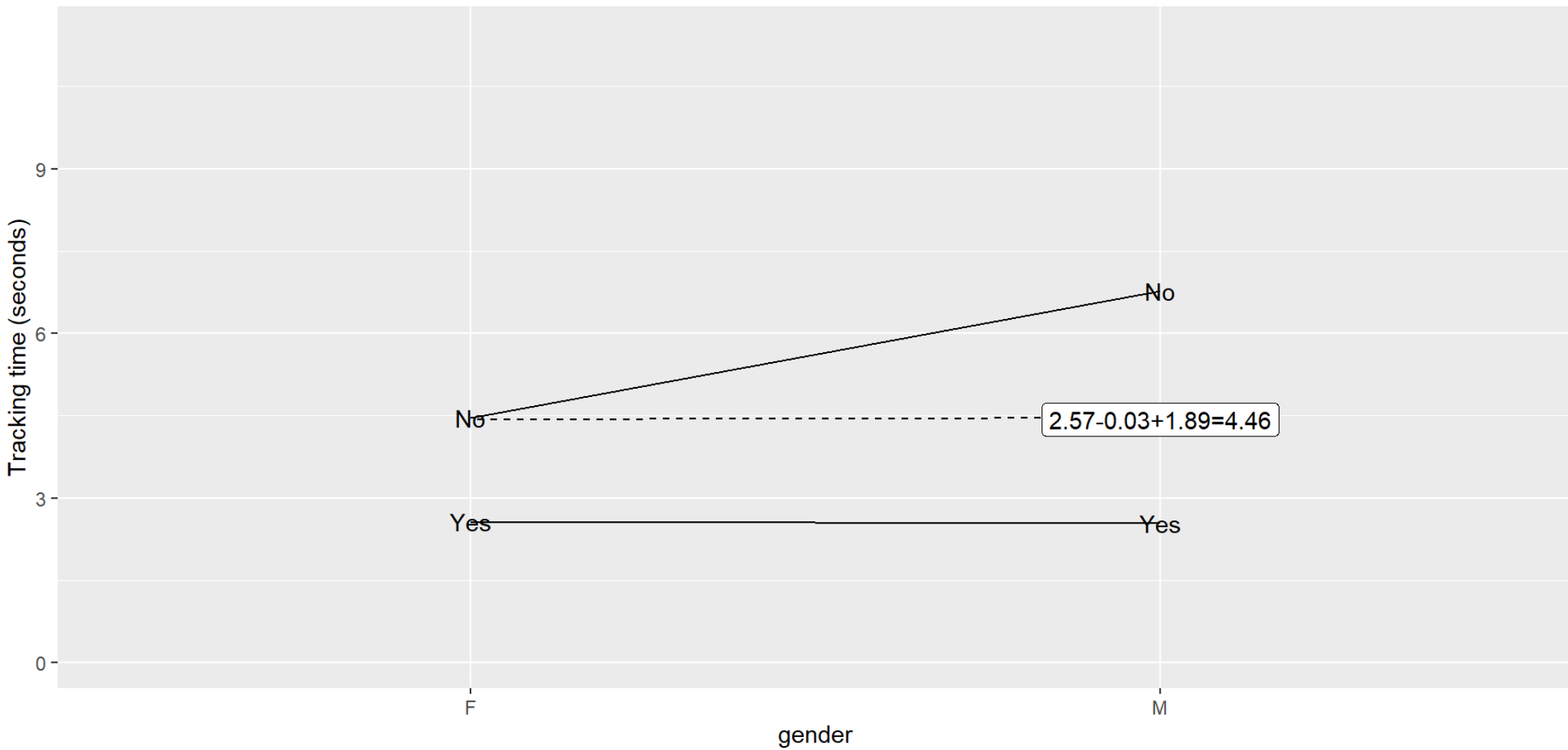


Simon, 2026-02-10

Speaker notes

Add the childNo term to the intercept to get the estimated average tracking time for female adults.

intercept + genderMale + childNo

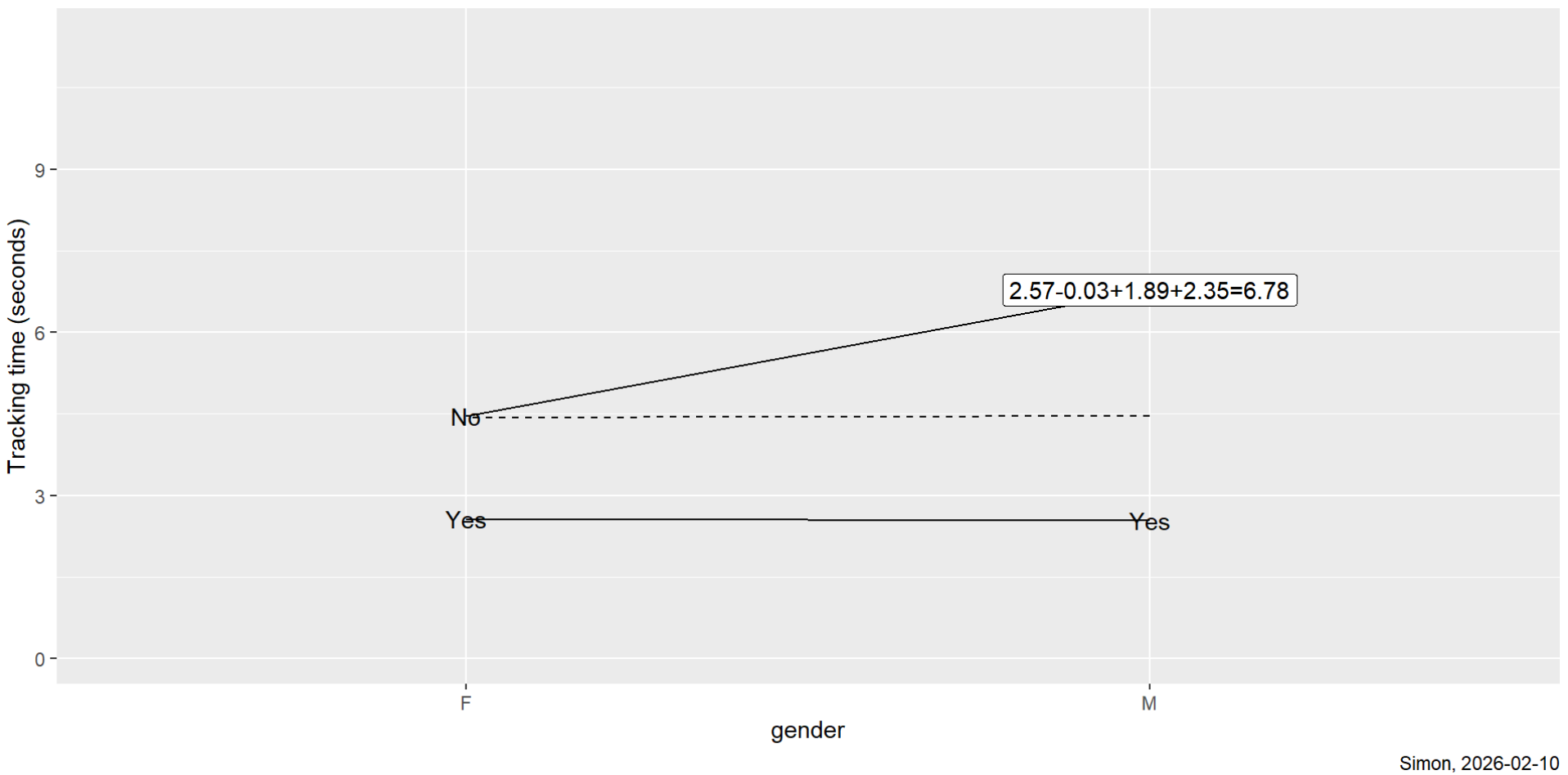


Simon, 2026-02-10

Speaker notes

If there were no interaction, then you would add the estimates of effect of genderMale and childNo to the estimated intercept get the estimated average time for adult males.

intercept + genderMale + childNo + genderMale:childNo



Speaker notes

Because there is an interaction, you have to add the estimates of the effect of the interaction to everything else. Male and adulthood are super-additive, meaning that the impact of both combined is more than you would expect from just looking at the individual effects.

Live demo, Interactions in multi-factor analysis of variance

Break #2

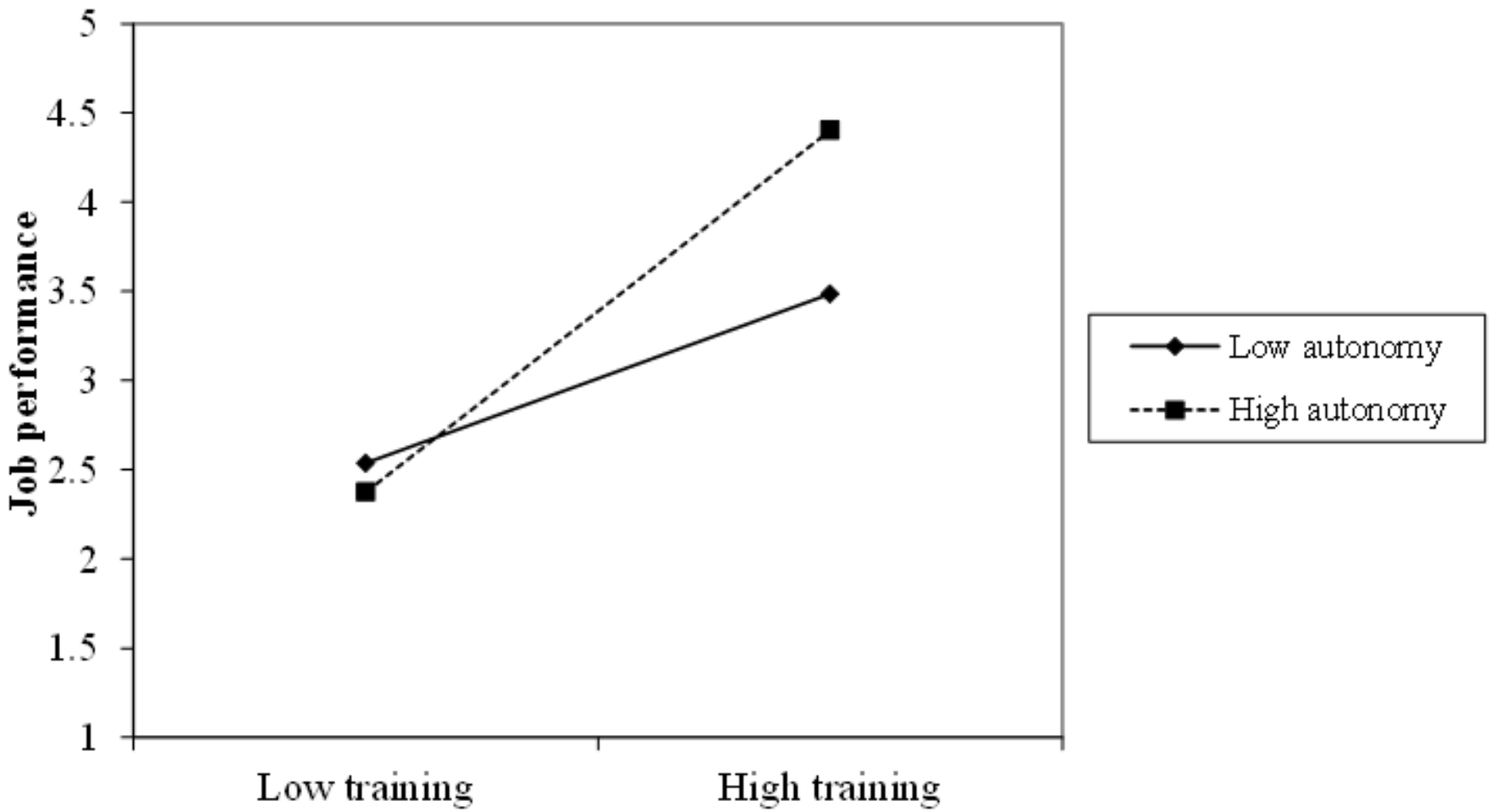
- What you have learned
  - Interactions in multi-factor analysis of variance
- What’s coming next
  - Other examples of interactions

Interaction of self-esteem and outcome

Speaker notes

Source: David M. Lane, Multi-Factor Between-Subjects Designs. Chapter XV, Section E of Available in html format.

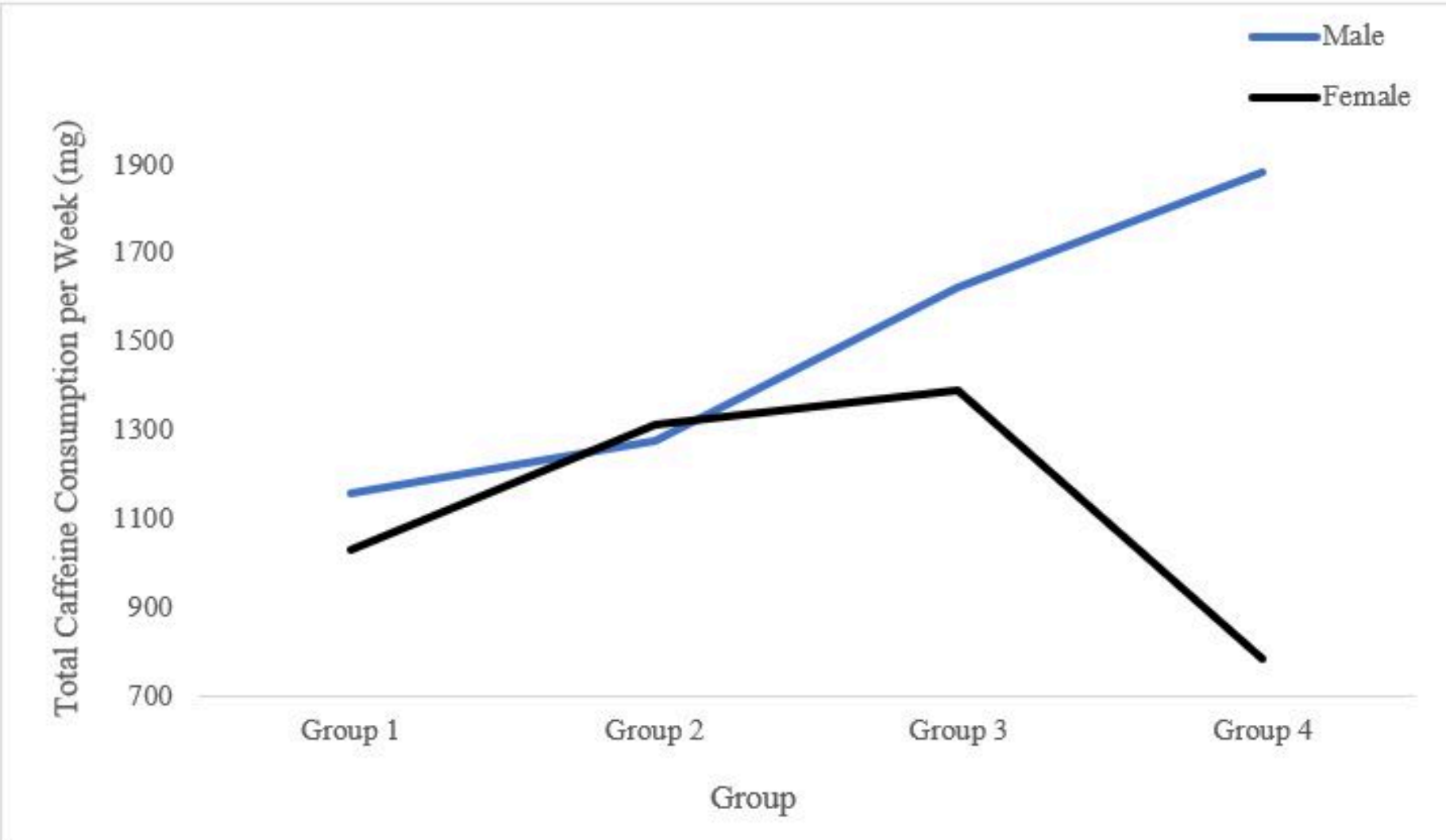
Interaction of training and autonomy



Speaker notes

Source: Wikipedia. Moderation (statistics). Available in html format.

Interaction of group and gender



Speaker notes

Source: Wikipedia. Analysis of covariance. Available in html format

Live demo, Other examples of interactions

Break #3

- What you have learned
  - Other examples of interactions
- What's coming next
  - Interactions in analysis of covariance

Interactions in analysis of covariance

- When you suspect an interaction
  - Graph your data
  - Draw a separate trend line for each group
  - Hope that the lines are mostly parallel
- Compute models with and without the interaction
  - Compare using the anova function
  - Center your continuous covariate (very important!)

Speaker notes

One of the assumptions of analysis of covariance is that there is no interaction between the covariate and the treatment/exposure variable. When the interaction is absent, you can totally remove the effect of the covariate.

An interaction changes things markedly. An interaction means that the effect of the covariate differs between the treatment group and the control group. You can no longer talk about removing the effect of the covariate by holding it constant. Instead, you have to recognize that the impact of the treatment varies, depending on the value of the covariate. A treatment that looks effective for some values of the covariate may be ineffective or even harmful for other values of the covariate.

If you suspect an interaction, first look at a graph. For the two levels of the treatment variable, draw a separate trend line associating the covariate with the outcome. If the lines are approximately parallel, that is a qualitative indication of no interaction.

A formal test of interaction requires you to fit a model with and without the interaction. Compare these two models using the anova function.

If you want to try to interpret the coefficients of the interaction model, you must (this is very important) center your covariate by subtracting the mean. Subtracting the mean places in the intercept in the heart of the data–no dangerous extrapolations needed.

Example: three exercise regimens

- Two exercise regimens, jogging, swimmin
- One control group: reading
  - Subjects tracked hours per week (mean = 2)
- Outcome variable, weight loss
  - Negative values represent weight gain

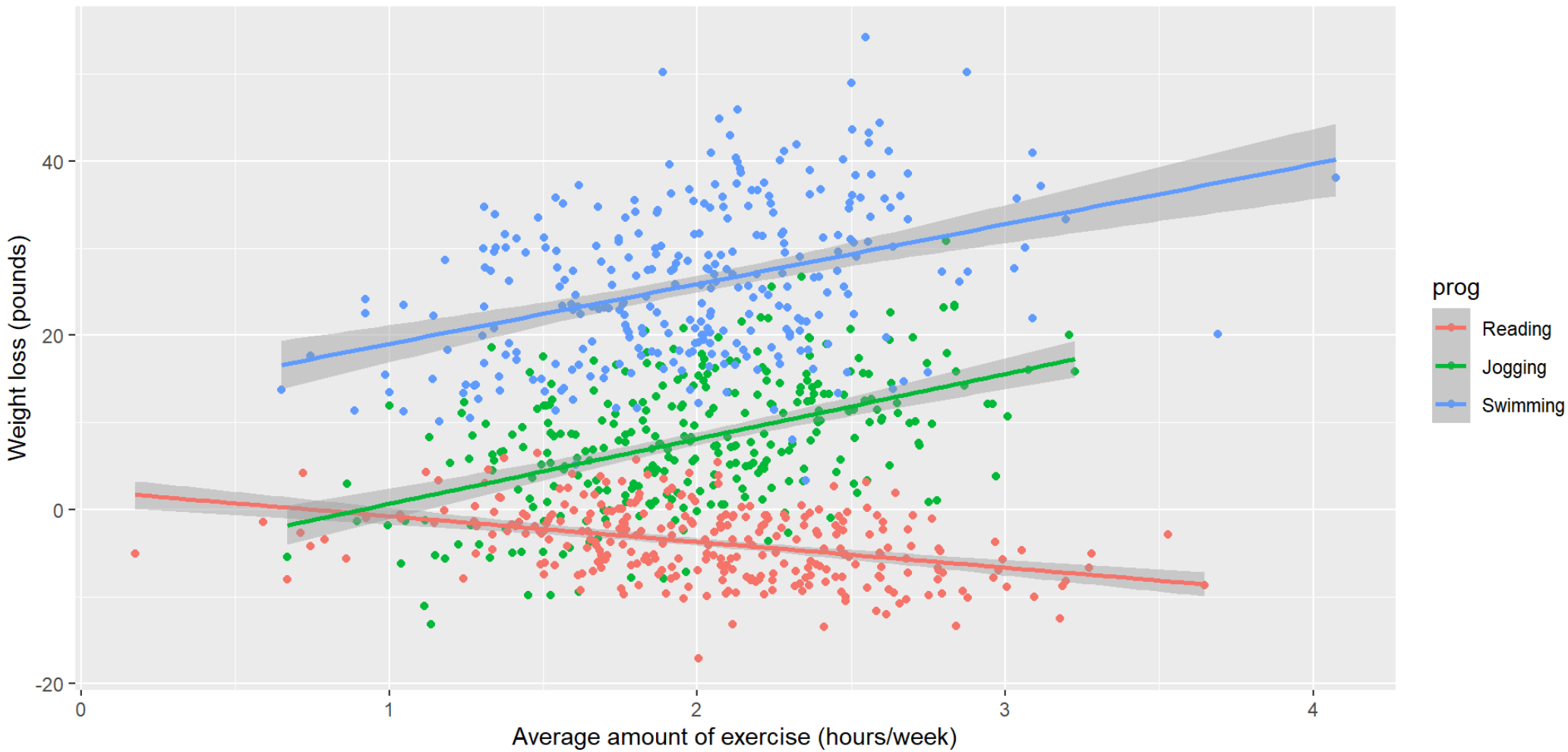
Speaker notes

Patients were put on two different exercise regimens, jogging or swimming, or in a control group that was asked to read for the same amount of time that the other two groups were supposed to spend exercising. You migt want to look at the effect of the three interventions after adjusting for the amount of time spent in each intervention.

The mean amount of time was 2 hours per week, so I computed a centerd version of hours by subtracing 2. negative values for this new variable meant that the subject spent less time on the intervention than the average subject. Positive values mean that the subject spent more time on the intervention than the average subject.

The intercept represents the estimated average weight loss for subjects with hours on the intervention equal to the overall average of 2. This is not an extrapolation. Most subejct spent somewhere between 1 and 4 hours on the intervention.

Visualizing the interaction



Steve Simon, 2026-02-16, CC0

Speaker notes

There is a very clear interaction. In the jogging and swimming groups, the more time that the subject spent on the exercise regiment, the more weight that they lost. The trend is in the opposite direction for the reading regimen. More time reading meant more weight GAIN. Clearly reading does no come with calories, but I suspect that the time spent reading crowded out other casual activities like walking the dog. More reading meant less dog walking and less dog walking meant more weight gain.

Testing the interaction

```
# A tibble: 2 × 7
  term                          df.residual    rss    df sumsq statistic p.value
  <chr>                               <dbl>  <dbl> <dbl> <dbl>     <dbl> <glue>
1 loss ~ prog + hours_centered          896 43115.    NA   NA       NA   <NA>
2 loss ~ prog + hours_centered…         894 37795.     2 5319.      62.9 p < 0.…
```

Speaker notes

The output does not quite fit on this slide, but you can see enough to recognize what is going on.

The F statistic is large and the p-value is small. This tells you that an interaction model does a statistically significantly better job of prediction than a model with no interaction.

Coefficients and interpretation

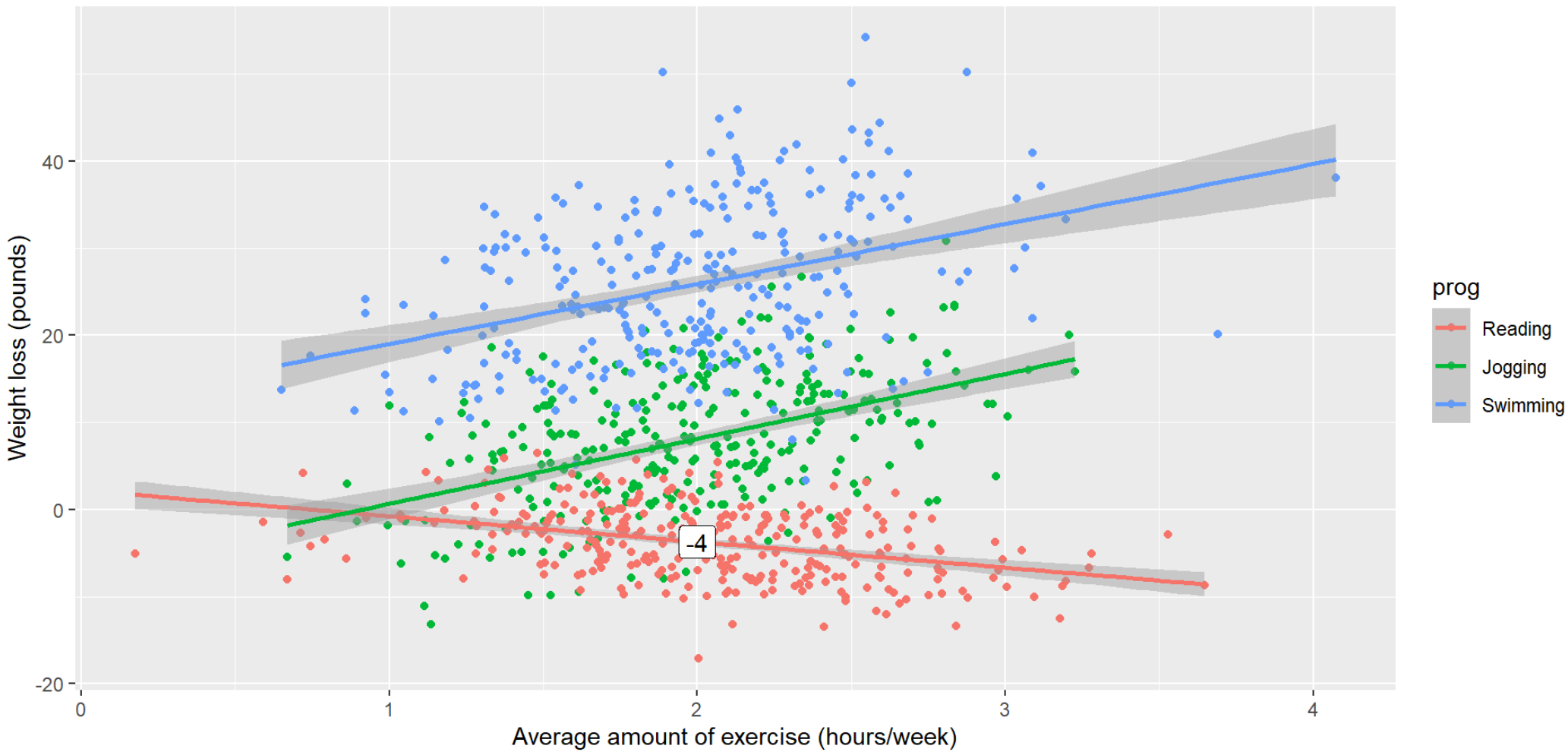
```
# A tibble: 6 × 5
  term                        estimate std.error statistic p.value
  <chr>                          <dbl>     <dbl>     <dbl> <glue>
1 (Intercept)                    -3.70     0.376     -9.85 p < 0.001
2 progJogging                    11.8      0.531     22.3  p < 0.001
3 progSwimming                   29.6      0.531     55.7  p < 0.001
4 hours_centered                 -2.96     0.708     -4.18 p < 0.001
5 progJogging:hours_centered     10.4      1.07       9.71 p < 0.001
6 progSwimming:hours_centered     9.83     1.05       9.35 p < 0.001
```

Speaker notes

The estimates are a bit tricky to interpret.

Let me show them to you on the graph to make things clearer.

Interpreting the intercept (-4)

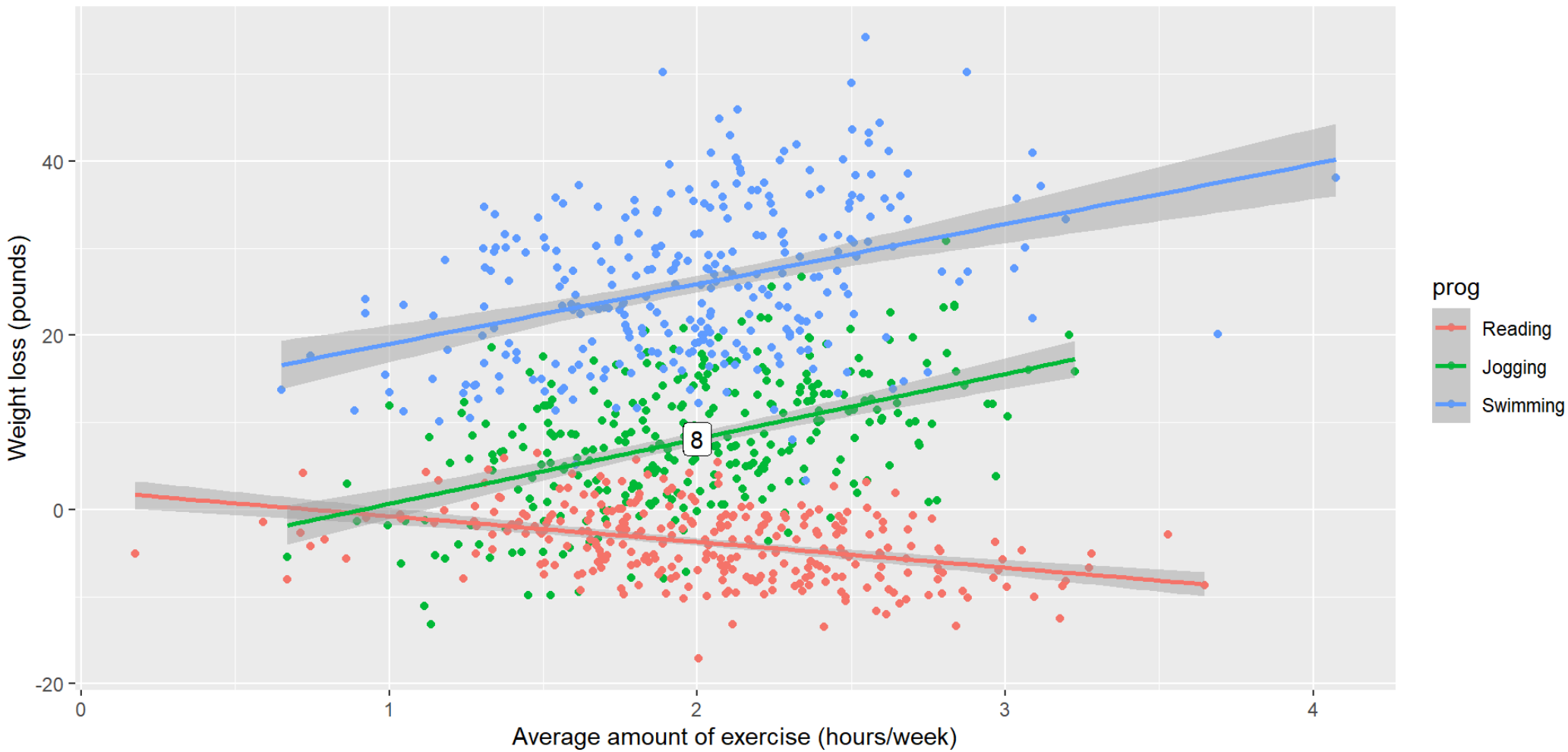


Steve Simon, 2026-02-16, CC0

Speaker notes

The intercept tells you that the estimated average weight loss is -4 pounds (actually a weight gain of 4 pounds) when the program is set to the reference value of reading and the number of hours_centered is zero (or equivalently, when the number of hours before centering is 2).

Interpreting progJogging (12)

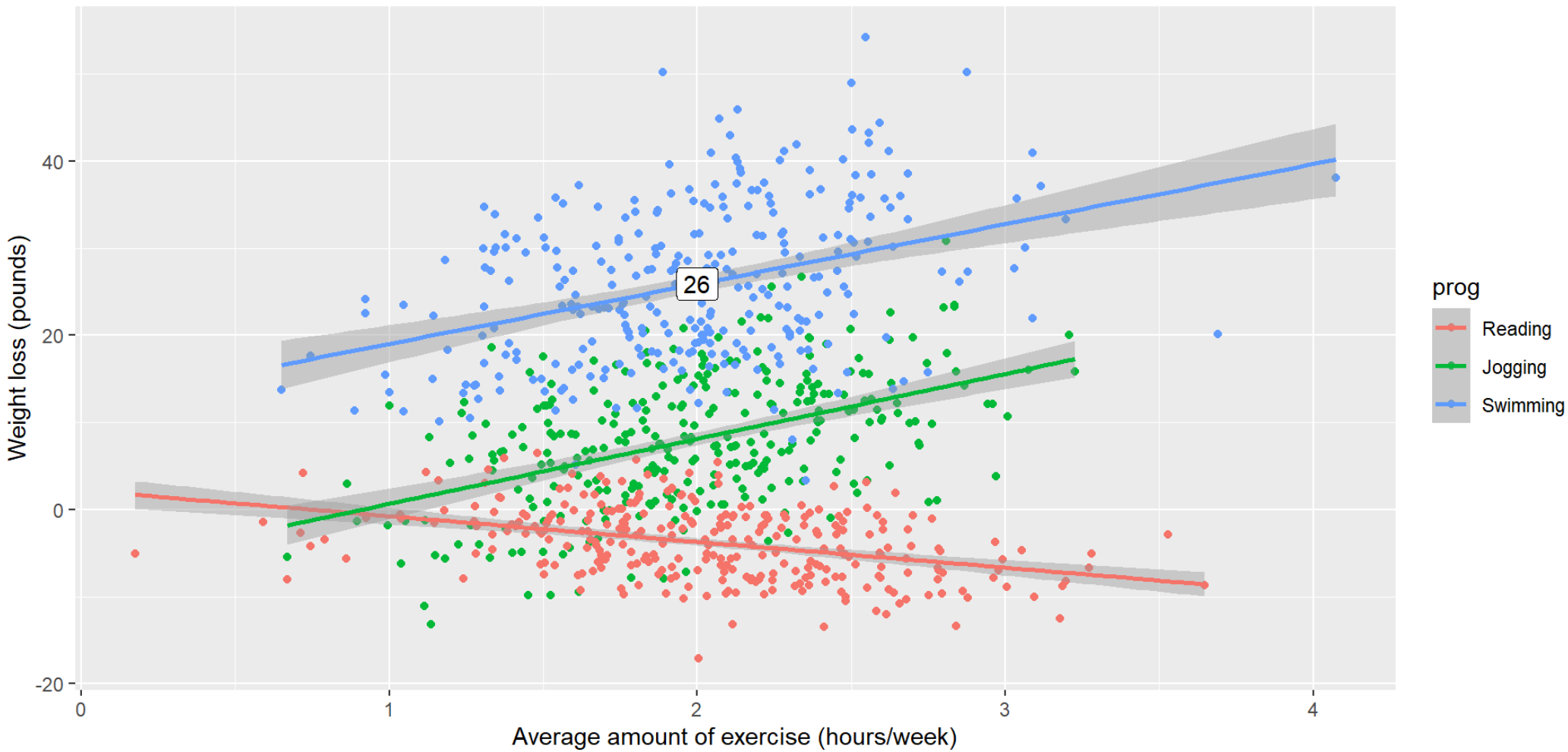


Steve Simon, 2026-02-16, CC0

Speaker notes

The progJogging term tells you that the estimated average weight loss improves by 12 pounds when you switch from reading to jogging and you hold the hours constant at 2. An improvement of 12 pounds means that you go from a weight gain of 4 pounds to a weight loss of 8 pounds when the time in the intervention is held constant at 2 hours per week.

Interpreting progSwimming (30)

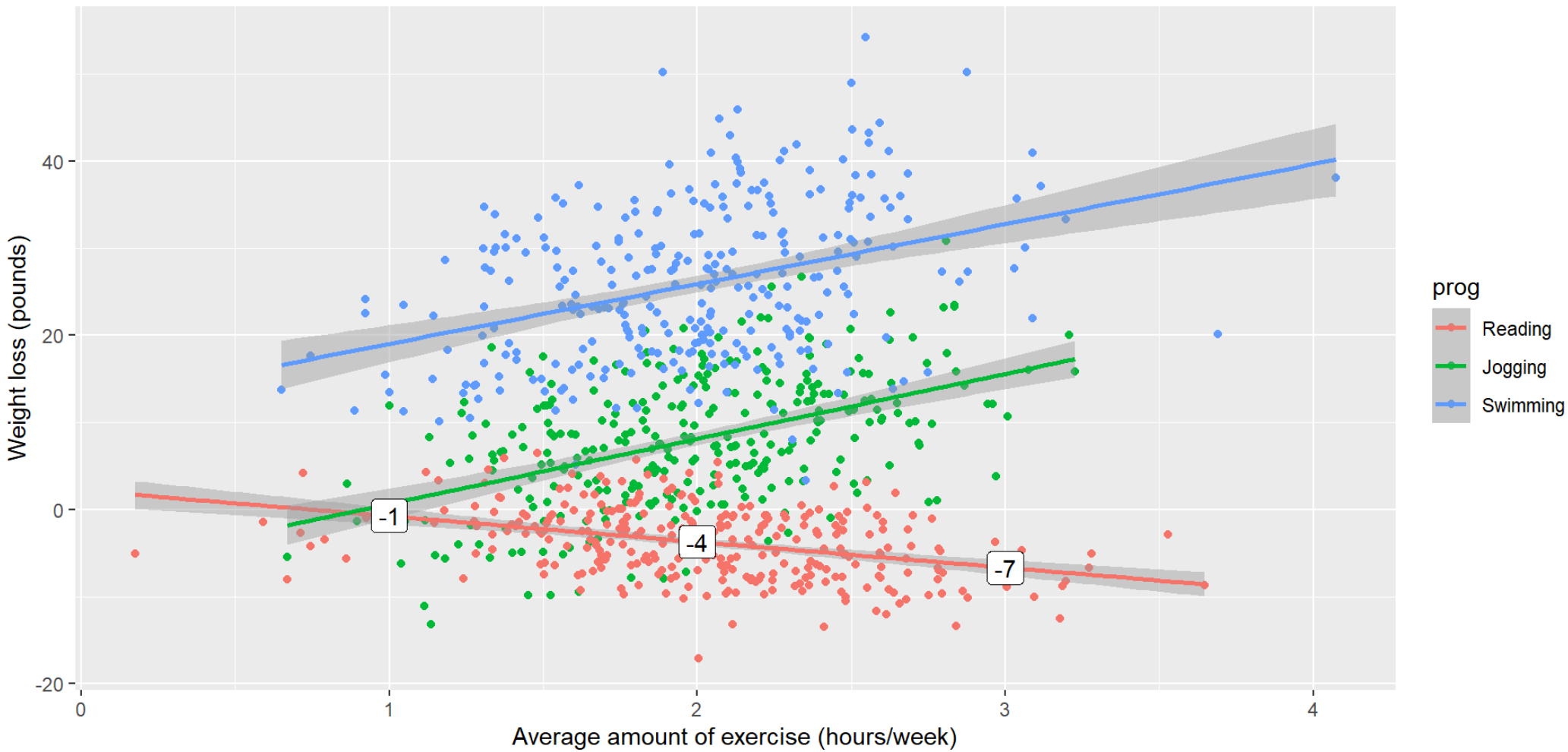


Steve Simon, 2026-02-16, CC0

Speaker notes

The progSwimming term tells you that the estimated average weight loss improves by 30 pounds when you switch from reading to swimming, holding teh hours constant at 2. An improvement of 30 pounds means that you go from a weight gain of 4 pounds to a weight loss of 26 pounds when the time in the intervention is held constant at 2 hours per week.

Interpreting hours_centered (-3)

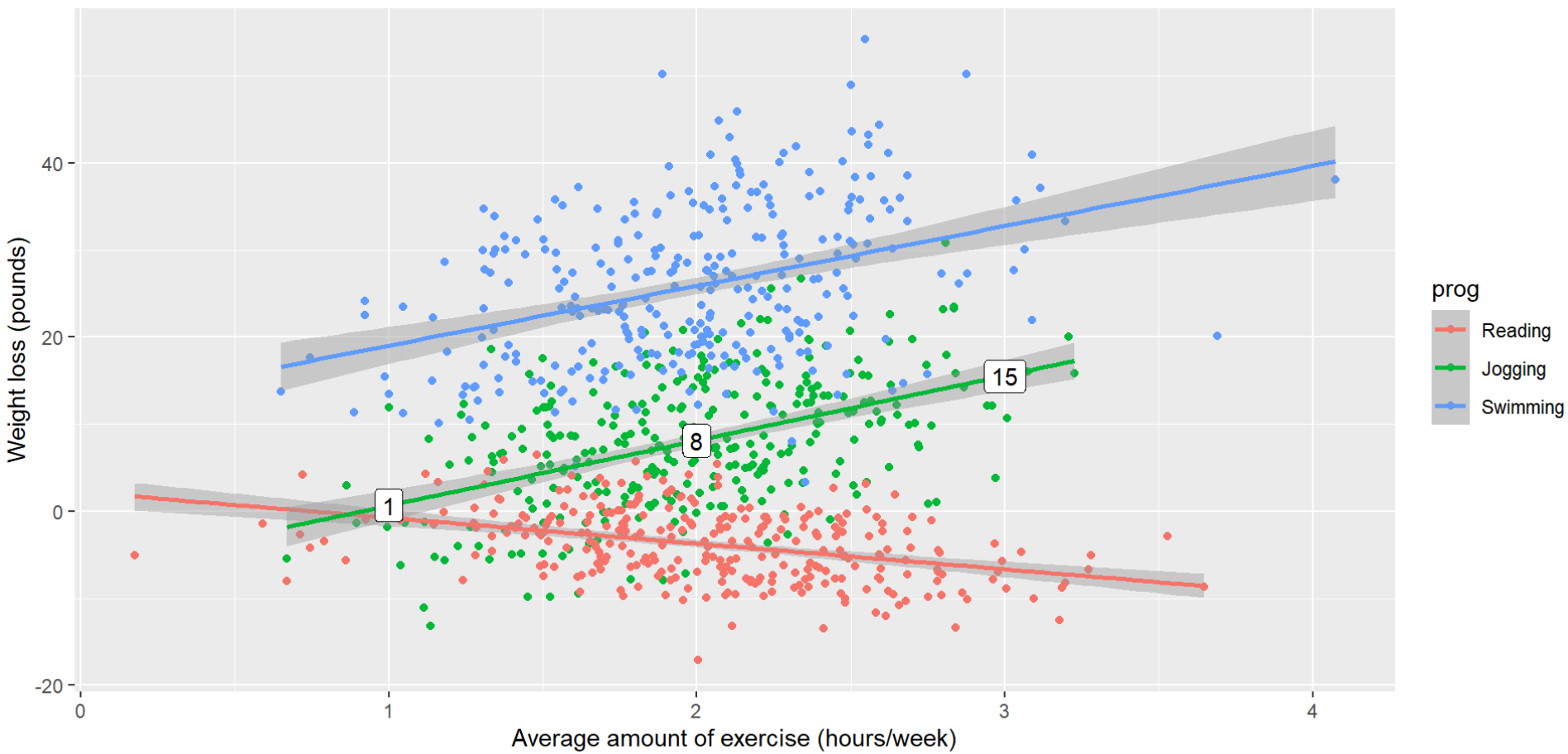


Steve Simon, 2026-02-16, CC0

Speaker notes

The hours_centered term tells you that you reduce your weight loss by 3 pounds (or increase your weight gain by 3 pounds) for change in one hour in the time spent in the intervention, holding the intervention at the reference level of reading. More reading means more weight gain. An average weight gain of 4 pounds for a volunteer with two hours of effort per week increases to an average weight gain of 7 pounds for a volunteer with three hours of effort per week. With one hour of effort per week, the average weight gain in the reading group is only 1 pound.

Interpreting progJogging:hours_centered: (10)

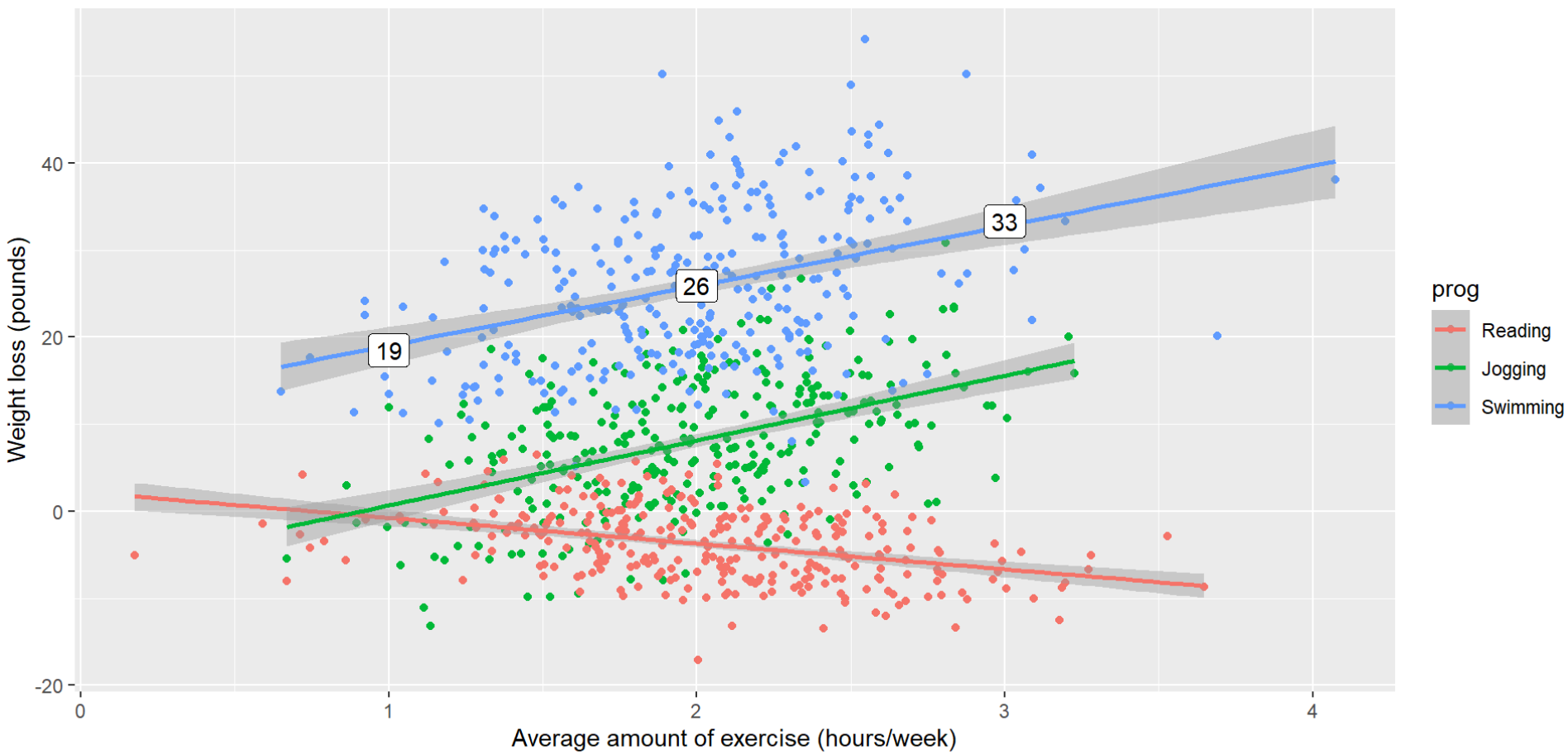


Steve Simon, 2026-02-16, CC0

Speaker notes

The progJogging by hours_centered interaction tells you that the change in weight loss is 10 pounds better for volunteers in the jogging group than the change that you see in the reading group after increasing the effort by one hour per week. This improvement means that the slope of -3 in the reading group changes to a slope of +7 in the jogging group. More jogging means more weight loss. An average weight loss of 8 pounds for a jogging volunteer with two hours of effort increases to an average weight loss of 15 pounds for a jogging volunteer with three hours of effort. You see a similar change but in the opposite direction for a jogging volunteer with only one hour per week of effort. Those volunteers see an average weight loss of only one pound.

Interpreting progSwimming:hours_centered: (10)



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Speaker notes

The progSwimming by hours_centered interaction tells you that the change in weight loss is 10 pounds better for volunteers in the swimming group than the change that you see in the reading group. This improvement means that the slope of -3 in the reading group changes to a slope of +7 in the swimming group. More swimming means more weight loss. An average weight loss of 26 pounds for a swimming volunteer with two hours of effort increases to an average weight loss of 33 pounds for a swimming volunteer with three hours of effort. The average weight loss for a swimming volunteer with only one hour per week effort is 19 pounds, not quite as big but still impressive.

General conclusions

- Jogging is better than reading, but only if you invest more than an hour per week.
- Swimming is better than jogging or reading.
- Extra time helps with the jogging and swimming interventions
- Extra time hurts with the reading intervention

Break #4

- What you have learned
  - Interactions in analysis of covariance
- What's coming next
  - Interactions in multiple linear regression

Interactions in multiple linear regression

- Center your continuous predictors

Summary

- What you have learned
  - Mathematical model of interactions
  - Interactions in multi-factor analysis of variance
  - Other examples of interactions
  - Interactions in analysis of covariance
  - Interactions in multiple linear regression